

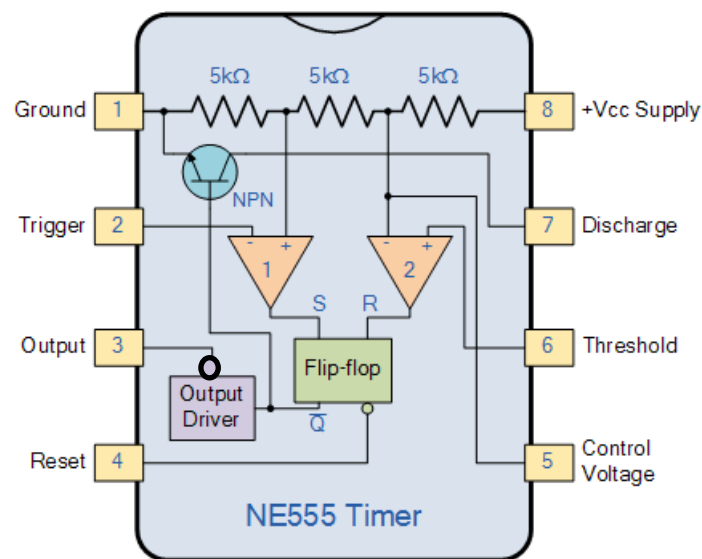
2.2 50% Duty Cycle PWM Generator Using NE555 Timer

OBJECTIVE AND OVERVIEW

In this task, we had to design and simulate a PWM (Pulse Width Modulation) generator circuit using the NE555 timer IC in LTspice. A 50% duty cycle stable square wave having a frequency in the range of 1 – 10 kHz was to be obtained at the output pin of the 555 timer. The same was observed along with the charging and discharging of the timing capacitor & the duty cycle, time of ON and time of OFF was measured. Usage of diodes with variable resistors or potentiometer to vary the duty cycle was also realized.

WORKING OF THE NE555 TIMER

There are 8 pins on the NE555 timer – Ground, Trigger, Output, Reset, Control Voltage, Threshold, Discharge, Supply voltage. Here is how it looks from the inside:



The working of this IC can be elaborated extensively, but to keep the discussion short and up to the point, we focus on a few key points:

1. **Resistors:** The three 5Ω resistors act as voltage dividers and thus divide 12V (say, $V_{cc} = 12V$) into 3 equal voltage drops. Hence, comparators 1 and 2 have their positive terminal and negative terminal fixed at 4V and 8V, respectively.

2. **Trigger and Threshold:** The trigger and threshold pins control the output of the two comparators. Normally, these two pins are externally connected together to some variable voltage. When this variable voltage is increased from 0 to 12V gradually, two things happen – when it crosses 4V ($V_{cc}/3$), the output of comparator 1 turns LOW from HIGH and when it crosses 8V ($2V_{cc}/3$) output of comparator 2 turns HIGH from LOW.

3. **Flip-Flop:** Accordingly, the flip flop takes on the values as shown below:

Voltage	Comparator 1	Comparator 2	Flip-Flop	Q	$\bar{Q}$	OUT
$\leq 4V$	HI	LO	Is SET	HI	LO	HI
$>4V \ \& \ <8V$	LO	LO	No Change	HI	LO	HI
$\geq 8V$	LO	HI	Is RESET	LO	HI	LO

4. **Discharge:** As soon as the flip-flop is reset, the NPN transistor turns ON and the discharge pin gets grounded. In the set condition of the flip-flop the NPN transistor remains OFF and discharge pin is essentially open.

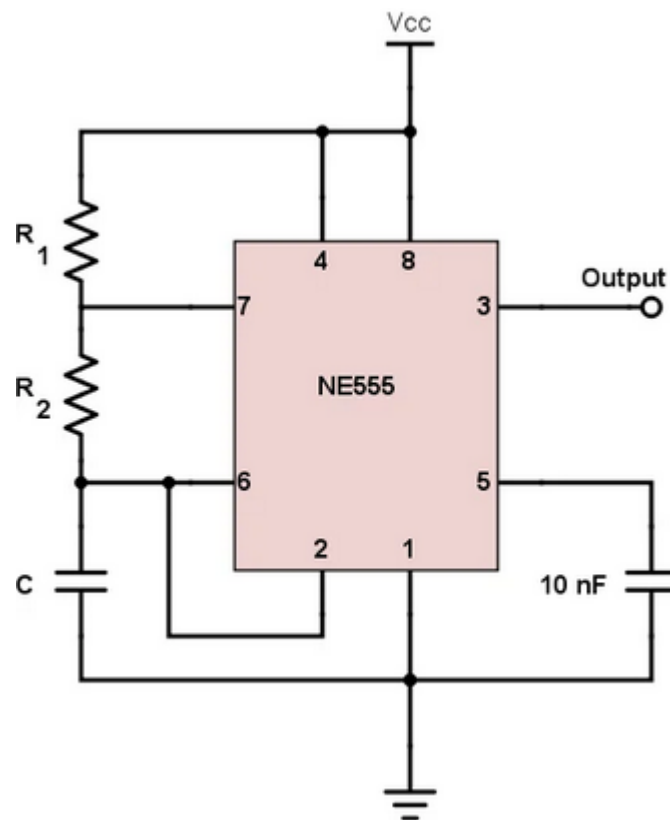
THEORY AND CONCEPTS

The duty cycle (in %) of the square waveform is given by:

$$Duty\ Cycle = \frac{ON\ time}{ON\ time + OFF\ time} \times 100\%$$

Using a capacitor connected between ground & trigger and threshold pins we can cause the capacitor to charge and discharge repeatedly between 4V and 8V without ever crossing these two extremities. This can most simply be seen when using the 555 timer in the astable mode of operation as shown below.

The following is an NE555 timer in the astable mode:



This usual configuration of the 555 timer cannot be used to obtain a duty cycle of 50%. It can be understood by looking at the flow of the process.

1. During the charge cycle, the current flows from V_{cc} to C through R_1 and R_2 until the voltage on the capacitor plate reaches $8V$ ($2V_{cc}/3$).
2. As soon as the voltage on the capacitor plate reaches $8V$ ($2V_{cc}/3$), the comparator 2 becomes HIGH, flip-flop resets, $\bar{Q}$ goes HIGH which turns on the NPN transistor and grounds the discharge pin, or the positive plate of the capacitor. Thus, the capacitor starts to discharge. The discharging of the capacitor occurs through R_2 only.

We therefore realize that the charging of the capacitor occurs through R_1 as well as R_2 but the discharging occurs only through R_2 . The equations of charging time and discharging time given below can be easily proved:

$$t_{charging} = (\ln 2) C (R_1 + R_2)$$

$$t_{discharging} = (\ln 2) C (R_2)$$

It is clear that the charging time will always be greater than the discharging time and hence the output (pin 3) of the 555 timer can never produce a 50% duty cycle waveform. We need something else.

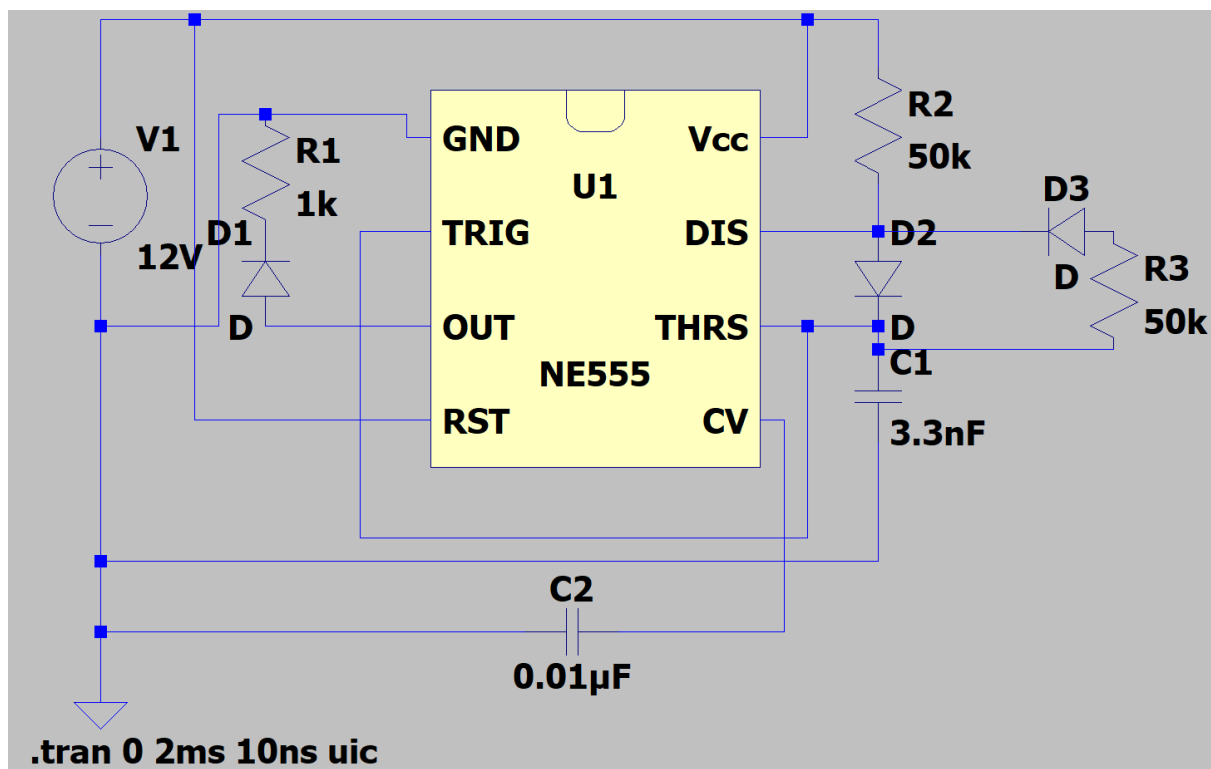
THE SOLUTION: 50% DUTY CYCLE

The solution to the discussed problem of traditional astable circuit is to use Diodes. Using diodes, we can restrict the path of current and force it to flow only through one of the two resistors such that the new equations become:

$$t_{charging} = (\ln 2) C_2 (R_2)$$

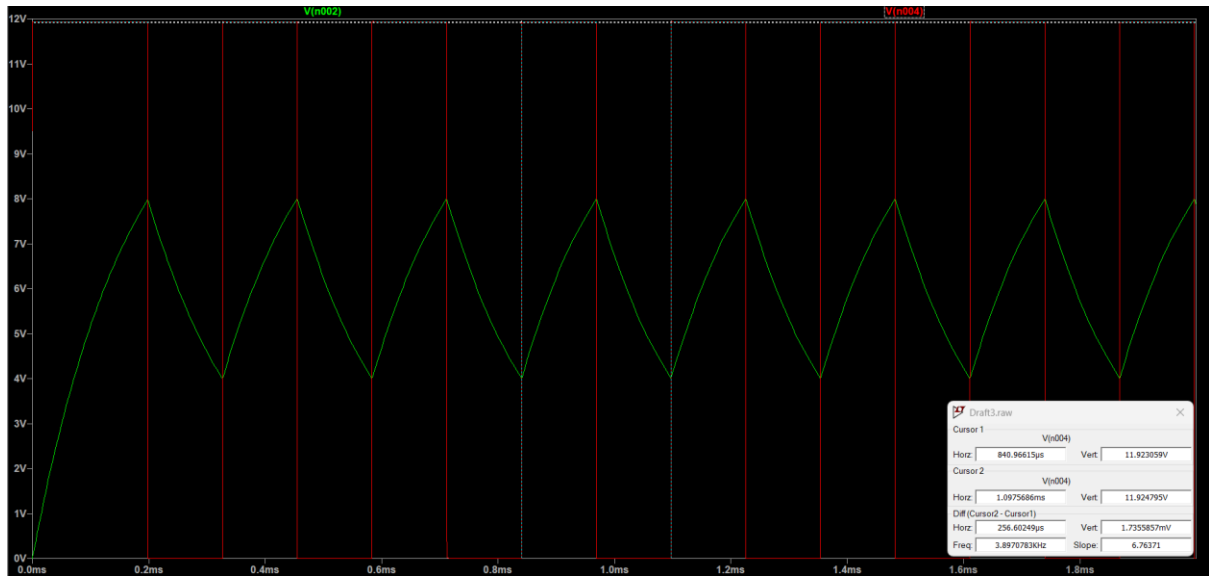
$$t_{discharging} = (\ln 2) C_2 (R_3)$$

After this, we make the resistor R_2 equal to R_1 so that we obtain 50% duty cycle. The circuit looks like:

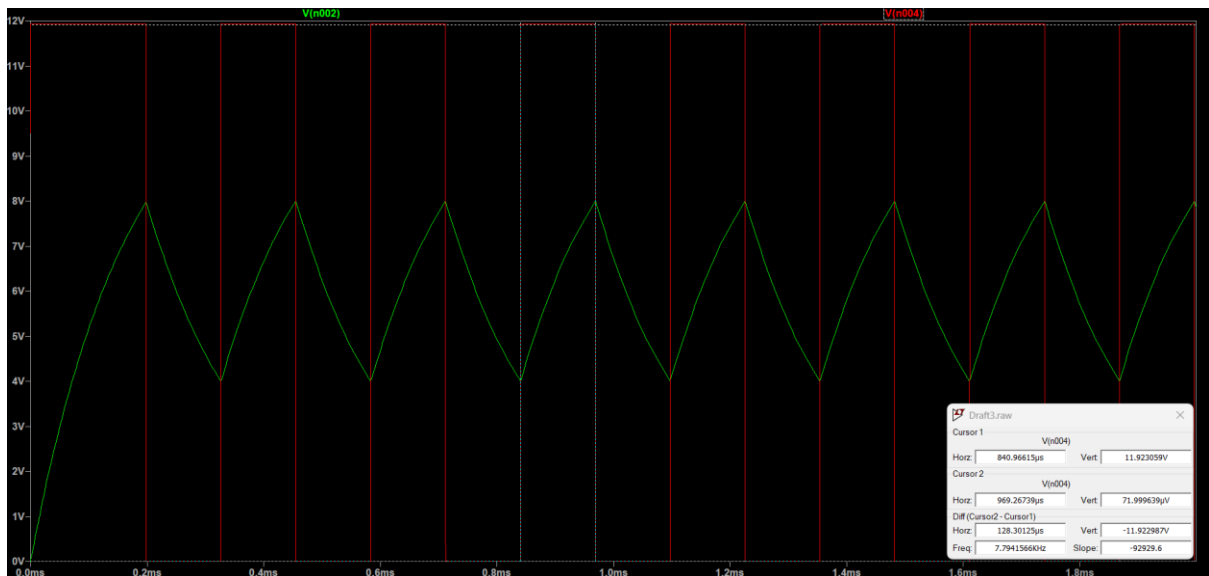


This way, while charging, the capacitor (3.3nF) charges through R_2 only and while discharging, it discharges only through R_3 . We can also vary the two resistors to change the duty cycle over a range as desired.

The corresponding voltage waveforms at output (in red) and of the plate of capacitor (in green):



The frequency as seen in the diagram above is about 3.90 kHz. Moreover, the charging time is:

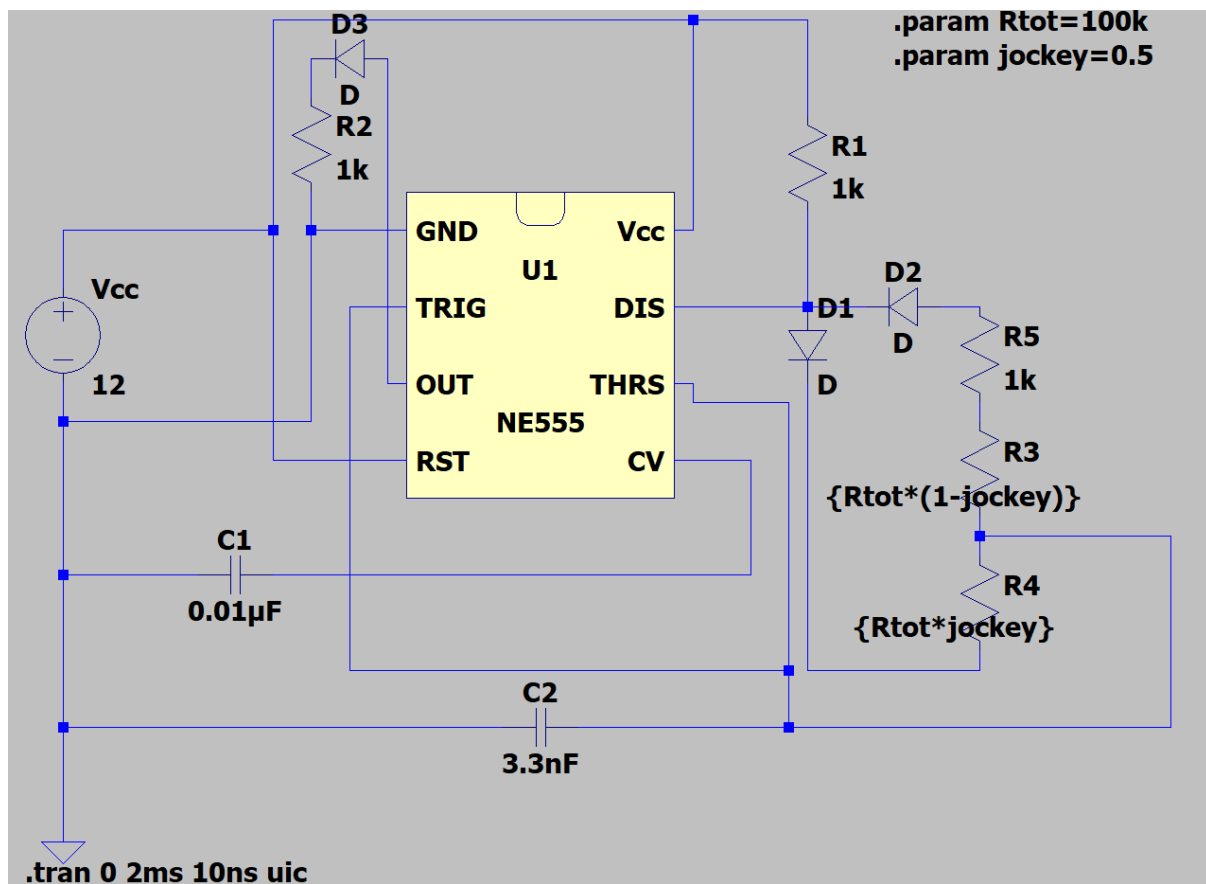


$t_{charging} = 128.30\mu s$, $t_{charging} + t_{discharging} = 256.60\mu s \Rightarrow t_{discharging} = 128.30\mu s$ as seen in the image above.

$$\therefore \text{Duty Cycle} = \frac{128.30\mu s}{256.60\mu s} = 50\%$$

An interesting thing worth noting is that had we not added the diode D_3 to the circuit above, we would have obtained a duty cycle of approx. 52.88% which is more than 2% out of range of 50%. D_2 effectively decreases the charging potential by V_F (characteristic of diode), and D_3 effectively increases the discharge floor by the same V_F . These two equal and opposite distortions cancel each other out exactly, making the duty cycle 50% regardless of temperature, current or the specific V_F of the diode. The only requirement is that the diodes should be the same.

THE FINAL CIRCUIT DESIGN



COMPONENTS USED

- Voltage source ($V_{cc} = 12V$)
- Ground
- 5 Resistors ($R_1 = 1k\Omega$, $R_2 = 1k\Omega$, $R_3 = R_{tot} (1 - jockey)$, $R_4 = R_{tot} \times jockey$)
- 2 Capacitors ($C_1 = 0.01\mu F$, $C_2 = 3.3nF$)
- 3 Diodes (D_1 , D_2 , D_3)
- Connecting wires

METHOD TO VARY THE DUTY CYCLE

Above shown circuit is the final circuit I thought of. This circuit is different than the circuit we discussed till the previous section in the sense that it has a potentiometer whose jockey is the variable voltage plate of the capacitor. This might seem a small adjustment, but it has a huge advantage. The most important advantage is that even if we move the jockey, the frequency of the waveform does not change. The reason can be found by writing simple equations. By extending the idea of charging time and discharging time we can now write:

$$t_{charging} = (\ln 2) C_2 (R_1 + R_{tot} \times jockey)$$

$$t_{discharging} = (\ln 2) C_2 [R_5 + R_{tot} \times (1 - jockey)]$$

$\therefore \text{time period} = t_{charging} + t_{discharging} = (\ln 2) C (R_1 + R_{tot})$ which is constant for a fixed potentiometer and capacitor. This way the $\text{frequency } (f) = \frac{1}{\text{time period}}$ remains constant. In the earlier circuit discussed this was not the case. Since the time period of the waveform remains fixed, the charging time and discharging time change proportionally to the change given in the value of *jockey*.

Another important advantage is ease of operation. To vary the duty cycle, the user simply has to change the value of the parameter (param) of jockey to obtain proportional change in charging and discharging time.

CIRCUIT OPERATION

Let us begin our discussion from the point when the capacitor is already charged to 4V. That is, we will discuss the working of the capacitor in the oscillatory region (between 4V and 8V).

1. Charging phase (Output = HIGH, Discharge = OPEN):

In this phase the NPN transistor connected to the discharge pin of the 555 timer is in the OFF state so essentially the discharge pin is OPEN circuited internally. The flow of the current is- $V_{cc} \rightarrow R_1 \rightarrow D_1 \rightarrow R_4 \rightarrow C_2$. D_2 is reverse biased so R_3 plays no role. The capacitor voltage at the trigger/threshold pin increases from 4V to 8V and during this the 555 timer holds its HIGH state at the output pin.

As soon as the capacitor voltage reaches 8V, the comparator connected to the threshold pin switches its state to HIGH, which changes the flip-flop state and the transistor turns ON. This grounds the discharge pin internally and initiates the discharging phase.

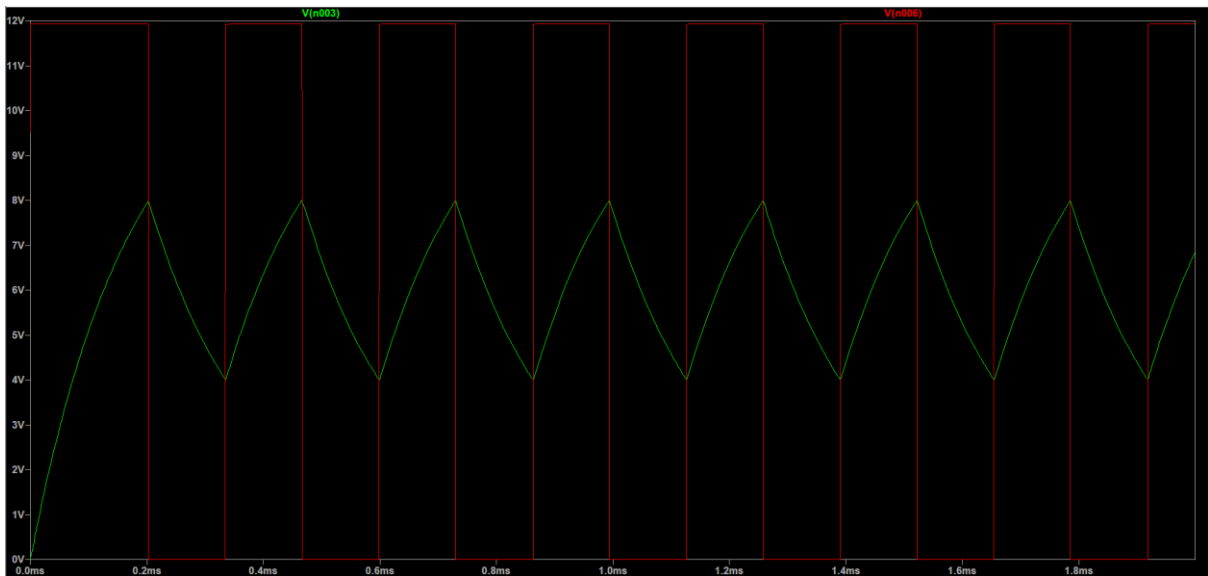
2. Discharging phase (Output = LOW, Discharge = GND):

The NPN transistor is turned ON in this phase and hence the discharge pin is grounded. The charged capacitor discharges gradually through the R_3 and R_5 according to the route- $C_2 \rightarrow R_3 \rightarrow R_5 \rightarrow D_2 \rightarrow Discharge \rightarrow GND$. Slowly the voltage of the capacitor comes down from 8V to 4V.

As soon as the capacitor voltage drops to just below 4V, the comparator connected to the trigger pin switches to HIGH which changes the flip-flop state and the transistor turns OFF. This opens the discharge pin internally and once again initiates the charging phase.

Hence the capacitor repeatedly oscillates between 4V and 8V, creating an approximate saw-tooth waveform while the output waveform is a square wave with apparently equal HIGH and LOW time.

The following image captures the Output waveform, capacitor charging and discharging waveform:



DESIGN EQUATIONS AND VALUES

$$\begin{aligned} \text{Duty Cycle} &= \frac{\text{ON time}}{\text{ON time} + \text{OFF time}} \times 100\% \\ &= \frac{t_{\text{charging}}}{t_{\text{charging}} + t_{\text{discharging}}} \\ &= \frac{(\ln 2) C_2 (R_1 + R_{\text{tot}} \times \text{jockey})}{(\ln 2) C_2 (R_1 + R_{\text{tot}} \times \text{jockey}) + (\ln 2) C_2 [R_5 + R_{\text{tot}} \times (1 - \text{jockey})]} \\ &= \frac{R_1 + R_{\text{tot}} \times \text{jockey}}{R_1 + R_5 + R_{\text{tot}}} \end{aligned}$$

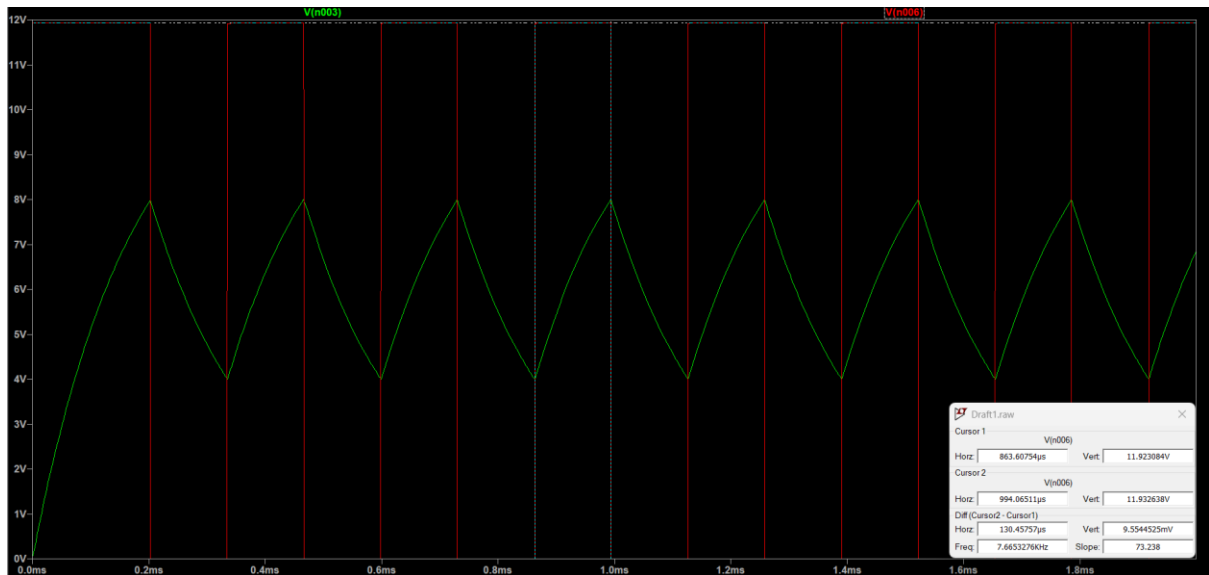
$$\begin{aligned}
 \text{Frequency} &= \frac{1}{t_{\text{charging}} + t_{\text{discharging}}} \\
 &= \frac{1}{(\ln 2) C_2 (R_1 + R_{\text{tot}} \times \text{jockey}) + (\ln 2) C_2 [R_5 + R_{\text{tot}} \times (1 - \text{jockey})]} \\
 &= \frac{1}{(\ln 2) C_2 (R_1 + R_5 + R_{\text{tot}})}
 \end{aligned}$$

Theoretically,

$$\text{Duty Cycle} = \frac{1k\Omega + 100k\Omega \times 0.5}{1k\Omega + 1k\Omega + 100k\Omega} = 50\%$$

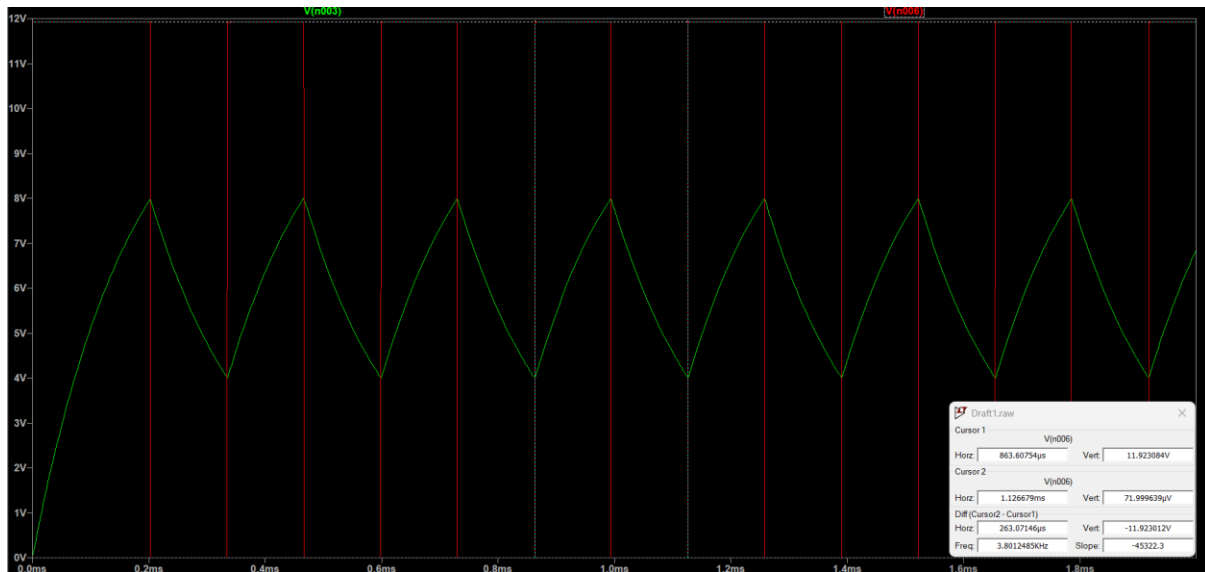
$$\text{Frequency} = \frac{1}{(\ln 2)(3.3 \times 10^{-9})(1k\Omega + 1k\Omega + 100k\Omega)} \cong 4.29\text{kHz}$$

For practical measurement of duty cycle and frequency, consider the following images of the capacitor and output waveforms:



From the above image, $t_{\text{charging}} = 130.458\mu\text{s}$ and from the image shown below, $\text{Time period} = t_{\text{charging}} + t_{\text{discharging}} = 263.071\mu\text{s}$ which gives

$$\text{Duty Cycle} = \frac{130.458\mu\text{s}}{263.071\mu\text{s}} \cong 49.60\%$$



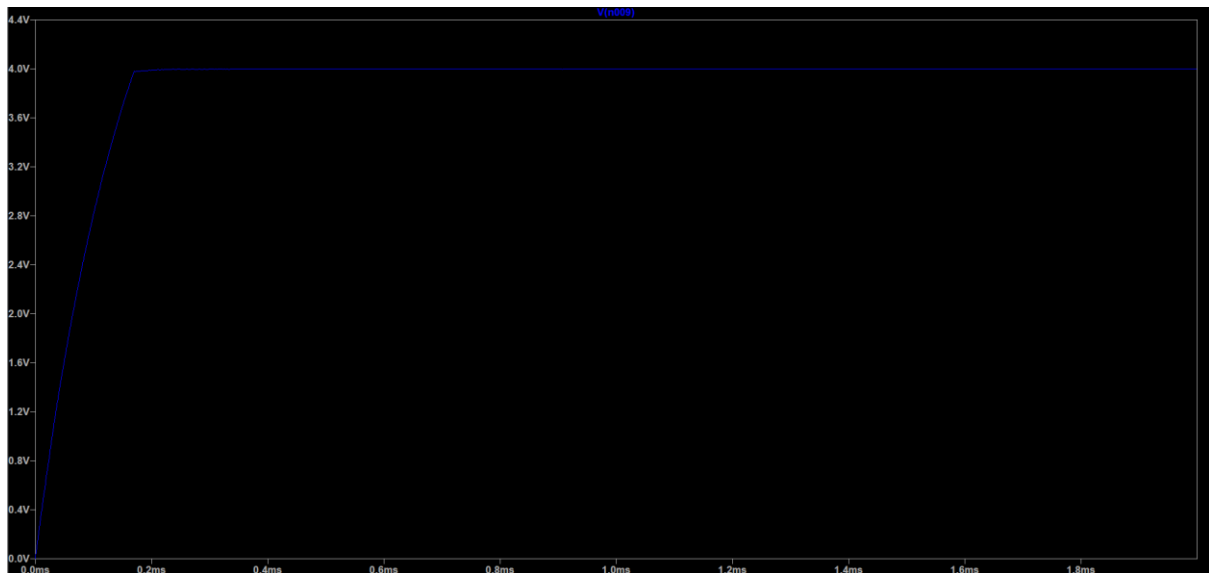
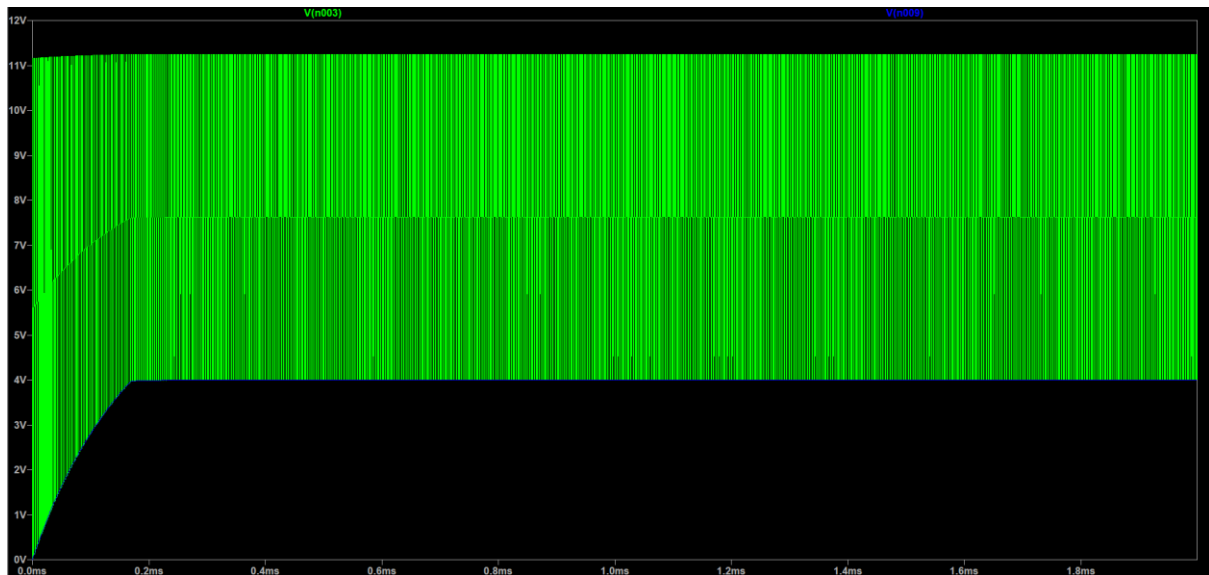
The frequency of the waveform can be easily measured to be about 3.801 kHz as seen in the image above.

The deviation of theoretical values and measured values occur due to the voltage drop across diodes (approx. 0.65V in LTspice) and saturation voltage (approx. 76mV in LTspice) of the NPN transistor at discharge pin of the 555 timer.

CHALLENGES FACED

1. I had initially connected threshold and trigger connecting wire not to the plate of the capacitor, but to the cathode of diode D_1 which was wrong. For this wrong connection I was getting a mushed up waveform at both the output of the 555 timer and the capacitor. I later figured out the reason as to why I was getting such a waveform. This was because as soon as the capacitor charged even for a small while, the discharge pin got immediately grounded because the threshold pin received a voltage much greater than 8V at the starting of the simulation itself. Moreover as soon as the capacitor discharged so that it had nearly 0 potential, the discharge pin again got immediately opened internally in the 555 timer because now the trigger pin fired (voltage at capacitor much less than 4V). This process continued at very short intervals of time and hence we got highly jagged waveform. Resolving this error took up a real lot of time.

See the images below of the error (output and capacitor voltage waveforms respectively):



2. Figuring out why we got a deviation between the measured values and calculated values was tricky.
3. The use of diode D_3 to compensate for the voltage drop across the other diode to make the duty cycle ratio 50%.